Fiscal Year:	FY 2022	**Task Last Updated:**	FY 06/07/2022
PI Name:	Turek, Fred W Ph.D.		
Project Title:	Effects of Spaceflight on Gastrointestinal Microbiota in Mice: Mechanisms and Impact on Multi-System Physiology		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Evanston	**State:**	IL
Zip Code:	60201-2919	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	06/01/2015	**End Date:**	05/31/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 5/31/2023 per NSSC information (Ed., 6/15/22) NOTE: End date changed to 5/31/2022 per NSSC information (Ed., 9/23/21) NOTE: End date changed to 5/31/2021 per F. Hernandez/ARC and NSSC information (Ed., 7/17/20) NOTE: End date changed to 5/31/2020 per F. Hernandez/ARC and NSSC information (Ed., 6/14/19) NOTE: End date changed to 5/31/2019 per NSSC information (Ed., 6/7/18)		
Key Personnel Changes/Previous PI:	June 2021 report: Co-Investigator Stefan Green has moved to Rush University from University of Illinois, Chicago (UIC).		

COI Name (Institution):	Green, Stefan Ph.D. (Rush University) Keshavarzian, Ali M.D. (Rush University Medical Center) Forsyth, Christopher Ph.D. (Rush University Medical Center) Vitaterna, Martha Ph.D. (Northwestern University) Jiang, Peng Ph.D. (Northwestern University)
Grant/Contract No.:	NNX15AL05G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	This proposal in response to NNH14ZTT002N NASA Research Announcement (NRA) “Research Opportunities for Flight Experiments in Space Biology” is intended to examine the potential role(s) of disruption of microbial communities in the gut (dysbiosis) in mammalian adjustment to the space environment. Many challenges to mammalian physiological homeostasis present in the space environment are known to create dysbiosis. Similarly, immune or inflammatory changes, as well as metabolic alterations, observed with time in spaceflight may be attributable in part to dysbiosis. A series of experiments in mice will be carried out to define the mechanisms by which time in space, diet, and host genotype interact to impact the composition of the gut microbiota, and how dysbiosis relates to gene expression (by RNA-seq) and physiology in serum, colon, ileum, spleen, liver, and fat as well as the sleep/wake cycle and feeding behavior. Extensive use of ground based studies will support the analysis and interpretation of spaceflight data. Five projects at three institutions will focus on different aspects. Project 1 (University of Illinois at Chicago, Principal Investigator (PI: SJ Green) will carry out metagenomic sequencing of microbiota from fecal samples and cecal contents (including from the tissue sharing opportunity); Projects 2 and 3 (Rush University Medical Center, PIs: A Keshavarzian and C Forsyth) will use gene expression, physiological and histologic measures of gastrointestinal and immune function, respectively; while Projects 4 and 5 (Northwestern University, PIs: FW Turek and MH Vitaterna) will focus on metabolism, sleep, and circadian behavioral organization. Dr. Turek (Northwestern) will provide Principal Investigator (PI) oversight. These studies are in alignment with NASA Research Announcement Research Emphases 1)“Spaceflight Omics Studies” and 3)“Understanding How Complex Organisms Adapt to Spaceflight.”
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal will examine the impact of the space environment on the population structure of the intestinal microbiota of mice, and how multiple physiological systems involving the gastrointestinal (GI), immune, metabolic, circadian, and sleep systems, known to be affected by the microbiota, are impacted by the space environment. While our overall working hypothesis is that the space environment and associated stressors will have a negative impact on the microbiota (i.e., dysbiosis), and that this in turn will negatively impact multiple physiological systems known to be adversely affected by spaceflight, our integrated analysis of gene expression and multiple tissue/organ specific biomarkers of tissue/organ injury will also allow us to uncover the adverse effects of spaceflight due or not due to dysbiosis. However, the impact of the proposed studies extends beyond understanding the mechanisms at play in the unique stresses of spaceflight. The proposed studies will elucidate mechanisms underlying interactions between GI, immune, metabolic, and sleep functions and specifically the role of the microbiota in these interactions. This knowledge has tremendous potential for guiding development of dysbiosis-targeted interventions for disorders in all of these systems.
Task Progress:	In the past year, data and sample collection from additional ground control groups has been completed. These 40 mice all were housed in the International Space Station Environment Simulator (ISSES) chambers in conditions corresponding to the original Rodent Research-7 (RR-7) flight, in NASA flight-like habitats. This control experiment was intended to enable a side-by-side comparison of the effects of the habitat light spectrum between blue-enriched (as was present in the ground controls of the original study) and white (as was present in the flight groups of the original study) on study outcomes. All animals successfully completed the study with final measurements at day 25 and day 75. Dissections were conducted with the Biospecimen Sharing Team. To avoid confounding effects of sequencing done at different times, RNA-seq analysis of tissue samples from the original and the new control groups will be done at the same time. Samples are being prepared for shipment to the NASA GeneLab for this analysis. Because of pronounced cohort effects on the gut microbiome, shotgun metagenomic sequencing of the microbiome samples from the original study has already been completed. In the past year, extensive analysis of this data has been completed and a manuscript is under preparation. Key findings include: 1) pronounced strain differences in the impact of the spaceflight on both alpha and beta diversity of the gut microbiome and in the changes in community structure over time; and 2) reproduction in RR-7 of the effects of spaceflight on the gut microbiome reported previously in RR-1 and STS-135 (Jiang et al., Microbiome, 2019). Analysis of video records in the past year has revealed strain differences in behavioral rhythms and in the use of the nestbox/hut within the habitat in flight, compared to ground habitats. In the coming year, we expect to complete analysis of our samples now that this can be done, including the new control groups. Reference: Jiang P, Green SJ, Chlipala GE, Turek FW, Vitaterna MH. "Reproducible changes in the gut microbiome suggest a shift in microbial and host metabolism during spaceflight." Microbiome. 2019 Aug 9;7(1):113. https:// ; PMID: 31399081; PMCID: PMC6689164
Bibliography Type:	Description: (Last Updated: 07/16/2025)
Dissertations and Theses	Kim SY. "Screening for contributing factors and countermeasures to the gut microbiome changes of spaceflight through bioinformatic analysis using STARMAPs." Senior Honors Thesis, Northwestern University Weinberg College of Arts and Sciences, May 2022. , May-2022